

ROKIB AL DHIN RAADH

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Independent AI researcher and founder specializing in autonomous, self-correcting neural architectures and multi-agent systems.

Publications & Technical Reports

R.A.D. Raadh. "ORMAS: Neural Architectural Transparency Enables Autonomous Self-Correction." 2026. Zenodo. DOI: [10.5281/zenodo.21730363](https://doi.org/10.5281/zenodo.21730363).

Research Experience

2024–pres.

Independent Machine Learning Researcher

ORMAS: Neural Architectural Transparency Enables Autonomous Self-Correction

PyTorch · 10,594 lines · 61 files · Codebase: anonymous.4open.science/r/ormas-EB73

- **Theory & Architecture:** Designed a three-signal training architecture: global backpropagation, a per-node local loss anchor (4,416-param bottleneck incapable of memorising noise), and a health-gated correction engine diagnosing 7 pathology types with expert immunity and selective rollback.
- Proved the first Input-to-State Stability (ISS) convergence guarantee for any self-correcting neural architecture; empirically validated by 83–93% correction magnitude decay over 200 epochs.
- **Catastrophic Shock Recovery:** Engineered ORMAS to autonomously recover to **80.3%** accuracy after its primary feature layer was zeroed mid-training — reclaiming 97% of lost performance. Standard CNNs collapse permanently to 10.0% (zero variance, $n=3$).
- Demonstrated recovery to **72.9%** (+62.9 pp gap) after all three convolutional layers were simultaneously zeroed — a scenario causing permanent, unrecoverable collapse in standard architectures.
- Under extreme weight perturbation ($\sigma=2.0$), ORMAS recovered to **62.1%** vs. 38.2% for a standard CNN (+52.1 pp) — the recovery gap widens monotonically with perturbation scale.
- **Generalization & Robustness:** Achieved **94.6%** shape memory retention during sequential task learning (ResNet-18: 47.3%); produced 58.8% zero-shot accuracy on novel unseen shape+colour combinations vs. 25% chance.
- Validated noise robustness across 6 regimes: asymmetric 40% corruption 80.6%; CIFAR-10N real annotator noise $1.4\times$ more stable than standard training; $<0.8\%$ accuracy variance across a $16\times$ hyperparameter range.
- **Scale & Reproducibility:** 383 controlled experiments across 4 architectures; 340 million telemetry entries, every correction causally tagged to pathology, node, and epoch — single RTX 3090.

Engineering

2023–pres.

Sole Architect & Engineer

OXIMO — Cognitive Multi-Agent Operating System

Python · 40,933 lines production · 2,011 tests · Codebase: anonymous.4open.science/r/oximo-5C73

- Built autonomous multi-agent system with hierarchical task decomposition, on-demand role creation via self-hiring pipeline, Graph-of-Thoughts planner, and 3-tier cognitive memory with temporal decay.
- Rebuilt from a **106,000-line monolith** — **72% line reduction** while preserving all **12 critical algorithms** with full audit trail. 47 feature flags across 17 implementation phases.
- Defense-in-depth safety: 7-check input cascade, PII redaction, injection pattern matching, and behavioral drift monitoring via EMA with Page-Hinkley change detection.

Ventures & Deployment

2024–pres.	Founder <i>Black Bloxie LTD — UK Registered Holding Company</i> A private equity structure deploying OXIMO as a scalable replacement for human operations across 4 verticals. • +1,300% revenue recovery & 99% cost reduction: Replaced human-operated workflows and expensive monolithic API calls (e.g., GPT-4) with an orchestrated cascade of six specialized, smaller AI models. Achieved equivalent output quality at 1% of the cost. • Vertical 1 (E-Commerce) — Causal Proof: 3-phase ablation over 1.5–2 years. Phase 1 (OXIMO deployed): 500+ customers, \$0 ad spend. Phase 2 (AI removed): revenue –91%, new customers –100%. Phase 3 (V3 re-injected): revenue +1,300%. Proved the system, not the market, drives the revenue. • Emergent AI-to-AI economy: 100% of acquisition driven autonomously by LLM referral traffic (ChatGPT, Bing, Gemini). Paid ads: zero. Human sales effort: zero. • Verticals 2–4 (Law, Software, Medical Research): Actively running. Full autonomous hiring, dynamic pricing, and self-modification validated by 2,011 tests; static boundaries currently held pending AI accountability regulation. • Registered at 17 from Bangladesh; navigated UK company formation, AML compliance, and cross-border operations independently.
2020–2024	Independent Ventures (5) <i>Automated Sales · AI Development · Market Tools</i> • Iterative testing across 5 independent ventures drove the architectural constraints and R&D foundation for OXIMO.

Technical Skills

ML / AI	PyTorch, PCGrad gradient surgery, ISS convergence theory, noise-robust training, multi-agent orchestration
Engineering	Python (advanced), C++, LaTeX, Docker, FastAPI, SQLAlchemy, ChromaDB, Prometheus, React, TypeScript
Languages	English (fluent), Bengali (native)

Education

Fall 2027	Prospective Undergraduate Seeking admission to pursue research in AI safety, self-correcting neural architectures, and autonomous systems.
2025–2026	Self-Directed Coursework • Deep Learning Specialization (5 courses), DeepLearning.AI (Andrew Ng). <i>Completed.</i> • MITx AI Programs — 25 courses across AI foundations, precision medicine, sustainable energy, and transportation. <i>Certificates in progress.</i>

Open-Source & Archives

ORMAS	zenodo.org/records/21730363 (permanently archived)
OXIMO	anonymous.4open.science/r/oximo-5C73 (anonymous review)
Codebase	anonymous.4open.science/r/ormas-EB73 (ORMAS source code)
Data	drive.google.com/... (383 experiments)